

ChoiceLoop

Program:

```
import java.util.*;
class StudentInfo {
    private int id;
    private String name;
    private int marks;
    private String subject;

    public StudentInfo(int id, String name, int marks, String subject) {
        this.id = id;
        this.name = name;
        this.marks = marks;
        this.subject = subject;
    }

    public int getMarks() { return marks; }
    public int getId() { return id; }
    public String getName() { return name; }
    public String getSubject() { return subject; }

    public static void showStudInfo(StudentInfo s1, StudentInfo s2) throws StudIdMarksException {
        System.out.println("Student 1: " + s1.getName() + ", Marks: " + s1.getMarks());
        System.out.println("Student 2: " + s2.getName() + ", Marks: " + s2.getMarks());
    }
}

class PerformanceEvaluator {
    public String comparebothStudentMarks(StudentInfo s1, StudentInfo s2) {
        if (s1.getMarks() > s2.getMarks()) {
            return s1.getName() + " has higher marks than " + s2.getName();
        } else if (s1.getMarks() < s2.getMarks()) {
            return s2.getName() + " has higher marks than " + s1.getName();
        } else {
            return "Both students have equal marks";
        }
    }
}

class StudIdMarksException extends Exception {
    public StudIdMarksException(String message) {
        super(message);
    }
}

public class ChoiceLoop {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;
        StudentInfo std1 = new StudentInfo(1, "Abc", 55, "Physics");
        StudentInfo std2 = new StudentInfo(2, "Xyz", 65, "Physics");
        PerformanceEvaluator pe = new PerformanceEvaluator();
        do {
            System.out.println("\nPlease Make a Choice:");
            System.out.println("1> Compare Marks Of 2 Students");
            System.out.println("2> Student Profile View");
        } while (choice != 1 && choice != 2);
    }
}
```

```

System.out.println("3> Average Marks Of Students from StudentList");
System.out.println("4> Compare Two Student Profiles");
System.out.println("5> Quit");

System.out.print("\nEnter Your Choice: ");
choice = sc.nextInt();
if (choice < 1 || choice > 5) {
    System.out.println("Invalid Choice! Please choose between 1-5");
    continue; // Skip to next iteration
}
switch(choice) {
    case 1:
        System.out.println("Chosen Option 1: Compare Marks");
        String result = pe.comparebothStudentMarks(std1, std2);
        System.out.println(result);
        break;

    case 2:
        System.out.println("Chosen Option 2: Student Profile View");
        try {
            StudentInfo.showStudInfo(std1, std2);
        } catch (StudIdMarksException ex) {
            System.out.println("Exception Thrown -----> " + ex);
        }
        break;

    case 3:
        System.out.println("Chosen Option 3: Average Marks");
        double average = (std1.getMarks() + std2.getMarks()) / 2.0;
        System.out.println("Average marks: " + average);
        break;

    case 4:
        System.out.println("Chosen Option 4: Compare Profiles");
        if (std1.getMarks() == std2.getMarks() && std1.getSubject().equals(std2.getSubject())) {
            System.out.println("Students have similar profiles");
        } else {
            System.out.println("Students have different profiles");
        }
        break;

    case 5:
        System.out.println("Thank you for using the system. Goodbye!");
        break;
}

} while(choice != 5);

sc.close();
}
}

```

Output:

Please Make a Choice:

- 1> Compare Marks Of 2 Students
- 2> Student Profile View
- 3> Average Marks Of Students from StudentList
- 4> Compare Two Student Profiles
- 5> Quit

Enter Your Choice: 1

Chosen Option 1: Compare Marks

Xyz has higher marks than Abc

Please Make a Choice:

- 1> Compare Marks Of 2 Students
- 2> Student Profile View
- 3> Average Marks Of Students from StudentList
- 4> Compare Two Student Profiles
- 5> Quit

Enter Your Choice: 2

Chosen Option 2: Student Profile View

Student 1: Abc, Marks: 55

Student 2: Xyz, Marks: 65

Please Make a Choice:

- 1> Compare Marks Of 2 Students
- 2> Student Profile View
- 3> Average Marks Of Students from StudentList
- 4> Compare Two Student Profiles
- 5> Quit

Enter Your Choice: 3

Chosen Option 3: Average Marks

Average marks: 60.0

Please Make a Choice:

- 1> Compare Marks Of 2 Students
- 2> Student Profile View
- 3> Average Marks Of Students from StudentList
- 4> Compare Two Student Profiles
- 5> Quit

Enter Your Choice: 4

Chosen Option 4: Compare Profiles

Students have different profiles

Please Make a Choice:

- 1> Compare Marks Of 2 Students
- 2> Student Profile View
- 3> Average Marks Of Students from StudentList
- 4> Compare Two Student Profiles
- 5> Quit

Enter Your Choice

Thank you for using the system. Goodbye!

